

$$O-P \Rightarrow \tau_p \left(\frac{dy}{dt} \right) + y = k_p f(t)$$

$$f(0) = 0. \quad (\text{Given})$$

$$G(s) = y$$

$$y' = s(G(s))$$

$$[s(G(s))] (\tau_p) + G(s) = k_p (f(t))$$

$$G(s) = \frac{(k_p) (f(t))}{s\tau_p + 1}$$

∴ transfer function is $\frac{k_p}{s\tau_p + 1}$